

Ratnesh Yadav, founders of Husk Power Systems (HPS), to do business from biomass, a power generation technology which uses anything from agri-residues to wood chips or even coconut shells. HPS's micro power plants, which burn rice husk to generate gas to run generators, supply electricity to 80 villages. The novel biomass technology converts rice husk into combustible gases, which then drive a generator to produce electricity at affordable rates. The per watt installation cost is Rs 50. Pandey, HPS's CEO, launched the company in August 2007 and set up the first plant in March 2008 which today supplies power mostly to off-grid locations in Bihar.

The social relevance is hard to miss, but Pandey and Yadav prefer to play it down: "We examined all the options and when we arrived at rice husk, Bihar emerged as the most resourceful place." What makes HPS's business model attractive is its low cost. One power

plant uses three quintals of rice husk per day to generate 32 KW of power, enough for 500 households. The rice husk costs Rs 60 per quintal. The production cost per plant per month is about Rs 20,000. So, HPS is able to sell electricity at Rs 40 per 15 watts. Pandey now wants to set up 10,000 power plants by 2014. This will be financed in part by the Rs 1.71 crore that it raised from US-based Acumen Fund and IFC's investment of \$1.25 million.

It is, however, solar power generation which is firing up entrepreneurial imagination in India and is the fastest growing cleantech sector in the country. No wonder, Delhi-based Green Infra, funded by IDFC, the largest equity player in cleantech, is straddling the shores of solar and biomass. Says Jain, "We already have 124 MW of operational wind power and are developing 24 MW of hydro power." According to James Abraham, Hemant Taneja and

Yogi Goswami, founders of SunBorne Energy, their choice of concentrated solar power model lies in the use of a huge number of mirrors which concentrate the radiation and then feed it into the steam turbine to produce power. SunBorne Energy intends to develop projects of over 1 GW in the next five-seven years, commission over 200 MW of solar power plants by 2014 and achieve grid parity in five to seven years.

Cost effectiveness and easier availability of finance today is also spawning start-ups in solar cleantech. Ask Inderpreet Wadhwa, a graduate from the University of California Berkeley who runs Azure Power, the country's first private sector company to implement a megawatt scale grid connected solar photovoltaic (PV) power plant. What makes Wadhwa's project unique is its cost-effectiveness. The 1 MW plant costs Rs 17-19 crore. Compared to this, public sector Bharat

▼ AZURE POWER

Entrepreneur: Inderpreet Wadhwa
Investor: Helion Advisers, IFC
IDEA: Megawatt-scale solar photovoltaic power plant, India's first private sector initiative

“**Cleantech is the precursor to a revolution in renewables, like what happened in telecom.**”

SUNIL JAIN, Founder, Green Infra



Petroleum Corporation Ltd is building a 1 MW PV power plant in Mohali for Rs 25 crore. Wadhwa argues that his costs will fall further once the prices of silicon and PV modules fall. "Besides, the Ministry of New and Renewable Energy allows for a further generation-based subsidy (varies in each state) which makes it a viable proposition," says Wadhwa.

The solar mission has indeed created the critical local demand, prompting Prasanth Sakhamuri to set up a solar module manufacturing unit for production of solar panels. HHV Solar's high efficiency solar panels can be used in commercial as well as residential applications. It currently manufactures a 300-watt solar panel, which, according